

Mitja Devetak

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Profile

Applied data scientist and research engineer for simulation-heavy, real-world data problems. Uses Python/JAX, Julia, C++, cloud/containerized workflows, and open-source scientific software to build calibrated models, network analytics, and decision-support systems.

Education

Complexity Science Hub Vienna	Vienna
<i>PhD in Complex Systems Science</i>	2024-present
University of Oxford	Oxford
<i>MSc Mathematical Modelling and Scientific Computing, Distinction</i>	2022-2023
University of Groningen	Groningen
<i>BSc Mathematics, cum laude; BA Political Philosophy, cum laude</i>	2019-2022

Experience

Freelance Quantitative Modelling Developer	Remote / Paris
<i>Various</i>	2024-present
- Built Python/JAX natural-disaster ABM workflows on AWS EC2 GPU/CPU jobs. - Developed C++ valuation modules for climate stress tests and Docker API integration. - Connected data, simulation, validation, and deployment in reproducible workflows.	
Supply Chain Intelligence Institute Austria	Vienna
<i>Researcher and PhD Candidate</i>	2024-present
- Built TIDES: one-to-one ABM of 35,954 ships moving among 1,653 ports. - Quantified exposure across maritime, energy, production-network, and firm datasets. - Validated simulation outputs and translated results into ministry briefings.	
Universite Paris 1 Pantheon-Sorbonne	Paris
<i>Research Assistant under Antoine Mandel</i>	2023-2024
- Developed maximum-entropy methods for global supply-chain reconstruction. - Built LLM-based socio-economic datasets and industry-aware firm networks. - Co-developed BeforeIT.jl for high-performance macroeconomic ABM.	

Skills

Data science:	statistical modelling, large-scale data analysis, calibration, validation, network modelling, simulation, decision-support analytics
Programming:	Python/JAX, Julia, C/C++, MATLAB, SQL; Docker, AWS EC2, Git, CI, Linux/Vim, LaTeX
Methods:	agent-based modelling, stochastic simulation, Markov models, maximum entropy, scenario analysis, numerical optimization

Additional

CFA Level I Candidate. RWRI 18: Risk Taking.